

Logic and questions

- When we pose a dichotomic question Q and we implicitly refer to a **specific interpretation** $\mathcal{I}$, the formalization in logic corresponds to checking whether $\mathcal{I} \models Q$. It follows that we are dealing with evaluating a formula (representing a Boolean questions) in an interpretation, and the possible answers are:
 - ▶ **yes**, if $\mathcal{I} \models Q$
 - ▶ **no**, if $\mathcal{I} \not\models Q$

Example: “In March, can you have sun while raining”? **yes**

- When we pose a dichotomic question Q and we refer to a **set of more than one interpretations**, denoted by a set of formulas Γ , the formalization in logic corresponds to checking whether $\Gamma \models Q$. It follows that we are dealing with logical implication, and the possible answers are:
 - ▶ **yes**, if $\Gamma \models Q$
 - ▶ **no**, if $\Gamma \models \neg Q$
 - ▶ **don't know**, if $\Gamma \not\models Q$ and $\Gamma \not\models \neg Q$

Example: “In March or August, can you have sun while raining”? **don't know** (depends on month)

Logical implication and formal reasoning

We discuss two basic forms of reasoning based on semantics, i.e., based on logical implication.

- **Deductive reasoning (deduction) via logical implication**

Check whether a formula A can be **deduced** from a set of formulas Γ , i.e., check whether

$$\Gamma \models A$$

- **Abductive reasoning (abduction) via logical implication**

Check whether a certain formula can be used to deduce another formula A in the context of a background knowledge represented by a set of formulas Γ , i.e., check whether there exists a formula B such that:

- ▶ $\Gamma \cup \{B\}$ is satisfiable, and
- ▶ $\Gamma \cup \{B\} \models A$

In other words, we are checking whether B can be **abduced** from A (or, for explaining A) in the context of Γ .

Logical implication and argumentation

We discuss two basic forms of argumentation and its formalization using logical implication.

- **Support**

To support a sentence S while arguing, one can exhibit a set of sentences (accepted as valid) that logically implies S

Γ supports S if $\Gamma \models S$

Example: “Paolo is adult” is supported by “Paolo has the driver’s license” and “license $\rightarrow$ adult”.

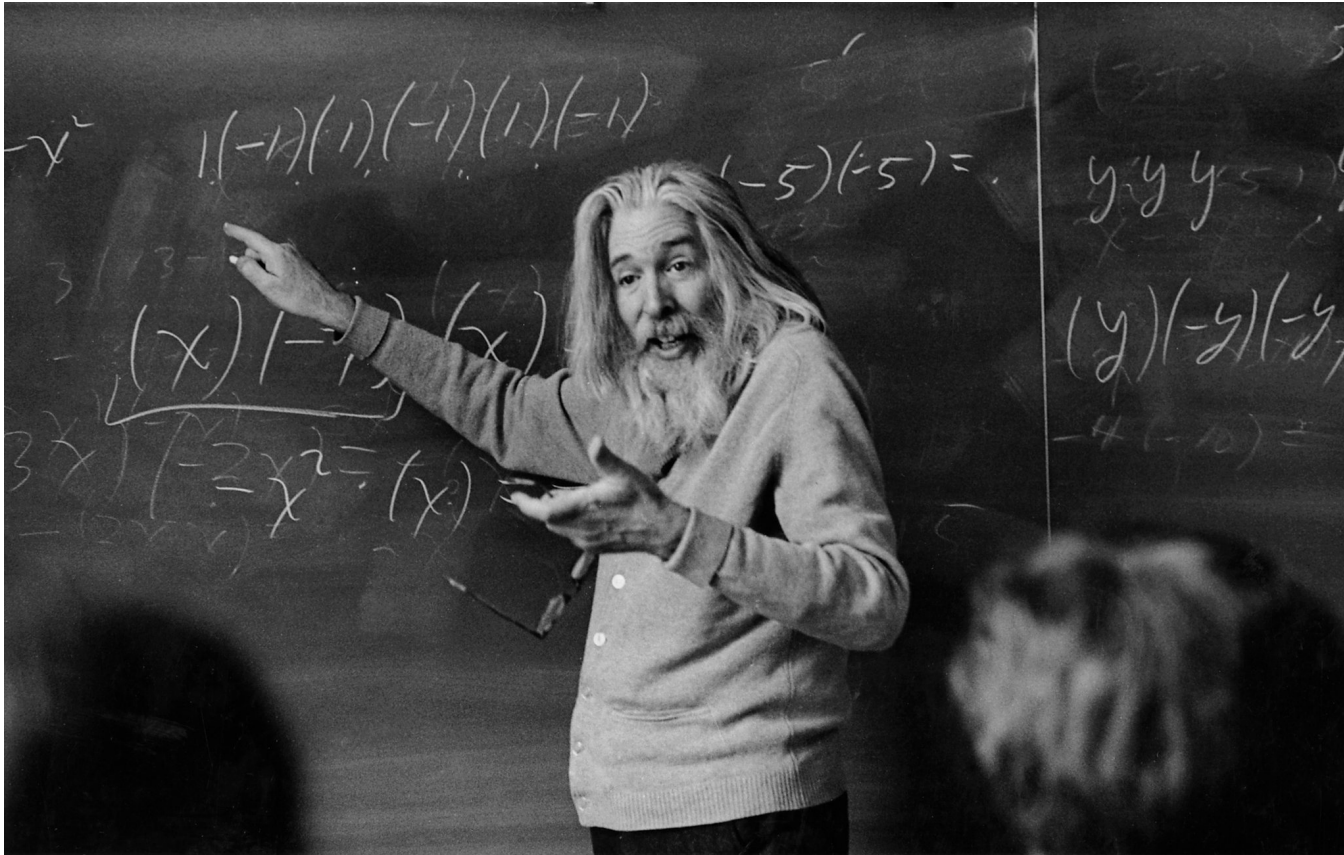
- **Refutation**

To refute a sentence S while arguing, one can exhibit a set of sentences (accepted as valid) that logically implies the negation of S

Γ refutes S if $\Gamma \models \neg S$

Example: “Paolo is adult” is refuted by “Paolo has the driver’s license”, “license $\rightarrow$ adult”, “adult $\rightarrow \neg$ underage”.

Raymond Smullyan's Tableaux Propositional Logic



Courtesy of Chiara Ghedini (FBK, Trento)

Outline of this lecture

- An introduction to Automated Reasoning with Analytic Tableaux;
- Today we will be looking into tableau methods for classical propositional logic (we'll discuss first-order tableaux later).
- **Analytic Tableaux** are a family of mechanical proof methods, developed for a variety of different logics. Tableaux are nice, because they are both easy to grasp for *humans* and easy to implement on *machines*.

How does it work?

The tableau method is a method for proving, in a mechanical manner, that a given set of formulas is **not satisfiable**. In particular, this allows us to perform automated *deduction*:

Given : set of premises Γ and conclusion ϕ

Task : prove $\Gamma \models \phi$

How? show $\Gamma \cup \neg\phi$ is not satisfiable (which is equivalent),
i.e. add the complement of the conclusion to the premises
and derive a contradiction (**refutation procedure**)

Reduce Logical Consequence to (un)Satisfiability

Theorem

$\Gamma \models \phi$ if and only if $\Gamma \cup \{\neg\phi\}$ is unsatisfiable

Proof.

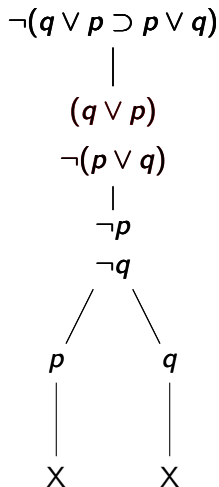
- $\Rightarrow$ Suppose that $\Gamma \models \phi$, this means that every interpretation $\mathcal{I}$ that satisfies Γ , it does satisfy ϕ , and therefore $\mathcal{I} \not\models \neg\phi$. This implies that there is no interpretations that satisfies together Γ and $\neg\phi$.
- $\Leftarrow$ Suppose that $\mathcal{I} \models \Gamma$, let us prove that $\mathcal{I} \models \phi$, Since $\Gamma \cup \{\neg\phi\}$ is not satisfiable, then $\mathcal{I} \not\models \neg\phi$ and therefore $\mathcal{I} \models \phi$.



Constructing Tableau Proofs

- **Data structure:** a proof is represented as a tableau a binary tree, the nodes of which are labelled with formulas.
- **Start:** we start by putting the premises and the negated conclusion into the root of an otherwise empty tableau.
- **Expansion:** we apply expansion rules to the formulas on the tree, thereby adding new formulas and splitting branches.
- **Closure:** we close branches that are obviously contradictory.
- **Success:** a proof is successful iff we can close all branches.

An example



Tableaux production rules for propositional logic

... for propositional connectives

α rules	$\frac{\phi \wedge \psi}{\phi}$ ψ	$\frac{\neg(\phi \vee \psi)}{\neg\phi}$ $\neg\psi$	$\frac{\neg\neg\phi}{\phi}$	$\frac{\neg(\phi \supset \psi)}{\phi}$ $\neg\psi$	
β rules	$\frac{\phi \vee \psi}{\phi \mid \psi}$	$\frac{\phi \supset \psi}{\neg\phi \mid \psi}$	$\frac{\neg(\phi \wedge \psi)}{\neg\phi \mid \neg\psi}$	$\frac{\phi \equiv \psi}{\phi \mid \neg\phi}$ $\psi \mid \neg\psi$	$\frac{\neg(\phi \equiv \psi)}{\phi \mid \neg\phi}$ $\neg\psi \mid \psi$

Expansion Rules of Propositional Tableau

α rules

$$\frac{\phi \wedge \psi}{\begin{array}{c} \phi \\ \psi \end{array}}$$

$$\frac{\neg(\phi \vee \psi)}{\begin{array}{c} \neg\phi \\ \neg\psi \end{array}}$$

$$\frac{\neg(\phi \supset \psi)}{\begin{array}{c} \phi \\ \neg\psi \end{array}}$$

$\neg\neg$ -Elimination

$$\frac{\neg\neg\phi}{\phi}$$

β rules

$$\frac{\phi \vee \psi}{\begin{array}{c|c} \phi & \psi \end{array}}$$

$$\frac{\neg(\phi \wedge \psi)}{\begin{array}{c|c} \neg\phi & \neg\psi \end{array}}$$

$$\frac{\phi \supset \psi}{\begin{array}{c|c} \neg\phi & \psi \end{array}}$$

Branch Closure

$$\frac{\begin{array}{c} \phi \\ \neg\phi \end{array}}{X}$$

Note: These are the standard (“Smullyan-style”) tableau rules.

We omit the rules for $\equiv$. We rewrite $\phi \equiv \psi$ as $(\phi \supset \psi) \wedge (\psi \supset \phi)$

Smullyans Uniform Notation

Two types of formulas: conjunctive (α) and disjunctive (β):

α	α_1	α_2	β	β_1	β_2
$\phi \wedge \psi$	ϕ	ψ	$\phi \vee \psi$	ϕ	ψ
$\neg(\phi \vee \psi)$	$\neg\phi$	$\neg\psi$	$\neg(\phi \wedge \psi)$	$\neg\phi$	$\neg\psi$
$\neg(\phi \supset \psi)$	ϕ	$\neg\psi$	$\phi \supset \psi$	$\neg\phi$	ψ

We can now state α and β rules as follows:

$$\frac{\alpha}{\alpha_1 \quad \alpha_2} \qquad \frac{\beta}{\beta_1 \mid \beta_2}$$

Note: α rules are also called **deterministic rules**. β rules are also called **splitting rules**.

Some definition for tableaux

Definition (Closed branch)

A **closed branch** is a branch which contains a formula and its negation.

Definition (Open branch)

An **open branch** is a branch which is not closed

Definition (Closed tableaux)

A tableaux is **closed** if all its branches are closed.

Definition

Let ϕ and Γ be a propositional formula and a finite set of propositional formulae, respectively. We write $\Gamma \vdash \phi$ to say that there exists a closed tableau for $\Gamma \cup \{\neg\phi\}$

Exercise

Show that the following are valid arguments:

- $\models ((P \supset Q) \supset P) \supset P$
- $P \supset (Q \wedge R), \neg Q \vee \neg R \models \neg P$

Solutions

